Cross-Scene Generalization Experiment Report: Tennis Ball Detection

1. Overview

This report summarizes the cross-scene generalization experiment for tennis ball detection using YOLOv26. The purpose of the experiment is to evaluate how different training dataset compositions affect detection performance across indoor, outdoor, and public-dataset scenes.

The experiment covers:

- 3 model scales: `yolo26n`, `yolo26s`, `yolo26m`
- 4 training sets: `R`, `RC1`, `RC2`, `RC1C2`
- 6 test sets: `R`, `C1`, `C2`, `RC1`, `RC2`, `RC1C2`
- 72 total evaluation runs

The main conclusion is that dataset diversity is more important than model size. Models trained on `RC1C2`, which combines Rainbow, Court1, and Court2, provide the most stable cross-scene performance.

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2. Dataset Description

2.1 Source Datasets

Dataset	Content	Test Size
R / Rainbow	Public tennis ball detection dataset with a relatively single visual domain	260
C1 / Court1	Outdoor real-court frames, annotated with a custom Python labeling tool	37
C2 / Court2	Indoor real-court frames	54

2.2 Training Set Compositions

Training Set	Content	Train	Val
R	Rainbow only	2072	259
RC1	Rainbow + Court1	2363	295
RC2	Rainbow + Court2	2496	312
RC1C2	Rainbow + Court1 + Court2	2787	348

2.3 Test Set Compositions

Test Set	Content	Test
R	Rainbow	260
C1	Court1	37
C2	Court2	54
RC1	Rainbow + Court1	297
RC2	Rainbow + Court2	314
RC1C2	Rainbow + Court1 + Court2	351

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3. Key Results

The following matrices are averaged across `yolo26n`, `yolo26s`, and `yolo26m`.

3.1 mAP50 Generalization Matrix

Train \ Test	R	C1	C2	RC1	RC2	RC1C2
R	0.9646	0.1049	0.8865	0.2456	0.9179	0.3850
RC1	0.9618	0.6242	0.9379	0.6745	0.9412	0.7396
RC2	0.9614	0.1324	0.9919	0.2646	0.9857	0.4462
RC1C2	0.9610	0.6210	0.9908	0.6728	0.9848	0.7603

3.2 mAP50-95 Generalization Matrix

Train \ Test	R	C1	C2	RC1	RC2	RC1C2
R	0.7131	0.0320	0.3948	0.1495	0.5155	0.2061
RC1	0.7021	0.2596	0.4880	0.3263	0.5704	0.3658
RC2	0.7119	0.0443	0.8609	0.1578	0.8165	0.3340
RC1C2	0.7048	0.2609	0.8548	0.3276	0.8109	0.4530

3.3 Best Overall Models

Rank	Model	Training Set	Avg mAP50	Avg mAP50-95	Avg Precision	Avg Recall
1	yolo26m	RC1C2	0.8364	0.5686	0.8125	0.8074
2	yolo26s	RC1C2	0.8333	0.5706	0.8030	0.8116

Rank	Model	Training Set	Avg mAP50	Avg mAP50-95	Avg Precision	Avg Recall
3	yolo26n	RC1C2	0.8255	0.5668	0.7880	0.8197
4	yolo26m	RC1	0.8242	0.4389	0.7831	0.7961
5	yolo26s	RC1	0.8178	0.4742	0.7937	0.7965

The top three models all use the `RC1C2` training set. This indicates that increasing dataset diversity has a stronger effect than simply increasing model scale.

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4. Key Findings

4.1 Court1 is the most difficult scene to generalize to

When trained only on Rainbow, the model achieves only 0.1049 average mAP50 on C1. Training on `RC2` also does not solve this issue, reaching only 0.1324 mAP50 on C1.

This shows that indoor Court2 data cannot replace outdoor Court1 data. If the final system needs to operate on outdoor courts, the training set must include Court1 or similar outdoor real-court samples.

4.2 Court2 is visually closer to Rainbow

The Rainbow-only model still achieves 0.8865 mAP50 on C2. After adding Court2, `RC2` reaches 0.9919 mAP50 and 0.8609 mAP50-95 on C2.

This suggests that C2 shares a stronger visual similarity with Rainbow than C1 does.

4.3 RC1C2 is the most stable training composition

`RC1C2` combines public data, outdoor real-court data, and indoor real-court data. It is not always the absolute best on every individual test set, but it provides the most balanced performance across all scenes.

This makes `RC1C2` the best training composition for general deployment and for the next-stage Ubuntu tracking test.

4.4 Model scale has limited impact compared with dataset diversity

The gap among `yolo26n_RC1C2`, `yolo26s_RC1C2`, and `yolo26m_RC1C2` is small:

- `yolo26n_RC1C2`: Avg mAP50 = 0.8255
- `yolo26s_RC1C2`: Avg mAP50 = 0.8333
- `yolo26m_RC1C2`: Avg mAP50 = 0.8364

The medium model is slightly better in average mAP50, but the small model provides almost the same accuracy with lower expected resource usage.

4.5 Localization accuracy remains a bottleneck

mAP50-95 is much lower than mAP50, especially for C1 and RC1. This means the models often detect the ball, but bounding box localization is still not precise enough.

This matters for downstream tasks such as:

- video tracking
- trajectory estimation
- depth-based 3D localization
- robot navigation and path planning

The next stage should therefore evaluate not only detection accuracy, but also tracking stability.

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5. Recommended Models for Ubuntu Tracking

The next stage will test model performance in Ubuntu video tracking. The recommended models are:

Model	Purpose	Weight Path
yolo26n_RC1C2	lightweight real-time baseline	`experiment/runs/yolo26n_RC1C2/weights/best.pt`
yolo26s_RC1C2	recommended balanced model	`experiment/runs/yolo26s_RC1C2/weights/best.pt`
yolo26m_RC1C2	accuracy upper bound	`experiment/runs/yolo26m_RC1C2/weights/best.pt`

5.1 yolo26n_RC1C2

This model has the smallest weight file, about 5.4 MB. It should provide the highest FPS and lowest GPU memory usage. It is suitable as the lightweight real-time baseline.

5.2 yolo26s_RC1C2

This is the recommended main model. It achieves near-medium accuracy while requiring fewer resources than `yolo26m`. It is the best first choice for Ubuntu tracking.

5.3 yolo26m_RC1C2

This model has the highest average mAP50 but also the largest weight file, about 44.0 MB. It should be used as the accuracy upper bound to determine whether the extra resource cost is worthwhile.

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6. Recommended Ubuntu Tracking Metrics

During the Ubuntu tracking test, each model should be evaluated on the same video input. The following metrics should be recorded:

Metric	Meaning
FPS	real-time video processing frame rate

Metric	Meaning
ms/frame	average per-frame processing latency
GPU memory	GPU memory usage, measured by `nvidia-smi`
GPU utilization	GPU workload
CPU usage	CPU resource cost
RAM usage	system memory usage
detection count	number of detected tennis balls
missed frames	frames where a visible ball is missed
false positives	non-ball objects detected as balls
tracking lost count	number of tracking interruptions
trajectory stability	smoothness and continuity of the tracked path

Final model selection should consider:

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detection accuracy + tracking stability + real-time performance + resource usage
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7. Next-Stage Testing Plan

Recommended testing order:

1. Test `yolo26s_RC1C2` first as the main balanced candidate.
2. Test `yolo26n_RC1C2` to measure the FPS improvement and accuracy trade-off.
3. Test `yolo26m_RC1C2` to measure whether the small accuracy gain is worth the extra computation.

If `yolo26s_RC1C2` satisfies FPS and tracking stability requirements, it should be selected as the final deployment model. If the Ubuntu device is resource-limited, use `yolo26n_RC1C2`. If maximum accuracy is required and GPU resources are sufficient, consider `yolo26m_RC1C2`.

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8. Summary

This stage confirms that:

- Dataset diversity is the dominant factor for cross-scene generalization.
- C1 outdoor data is necessary for outdoor court performance.
- RC1C2 is the most stable training composition.
- The three RC1C2 models are the best candidates for video tracking.
- `yolo26s_RC1C2` is currently the best balanced model for the next Ubuntu test.

The next stage should focus on real-time tracking experiments and compare FPS, resource usage, and tracking stability among `yolo26n_RC1C2`, `yolo26s_RC1C2`, and `yolo26m_RC1C2`.

9. Appendix: Full Evaluation Results per Model

The following tables are generated from `cross_eval_all_summary.csv`.

yolo26n

Train	Test	In-dist	Precision	Recall	mAP50	mAP50-95
R	R	Yes	0.9351	0.9248	0.9677	0.7368
R	C1	No	0.2795	0.0753	0.1356	0.0457
R	C2	No	0.9127	0.8969	0.9239	0.5164
R	RC1	No	0.6299	0.1645	0.2784	0.1724
R	RC2	No	0.9323	0.8941	0.9384	0.6038
R	RC1C2	No	0.7854	0.2944	0.4161	0.2503
RC1	R	No	0.9377	0.8885	0.9609	0.7135
RC1	C1	No	0.5638	0.6545	0.5953	0.2455
RC1	C2	No	0.9298	0.8881	0.9282	0.4694
RC1	RC1	Yes	0.6065	0.6714	0.6494	0.3141
RC1	RC2	No	0.9380	0.8793	0.9342	0.5635
RC1	RC1C2	No	0.6659	0.6916	0.7176	0.3525
RC2	R	No	0.9310	0.9115	0.9592	0.7171
RC2	C1	No	0.4901	0.0828	0.1370	0.0438
RC2	C2	No	0.9913	0.9805	0.9911	0.8661
RC2	RC1	No	0.6426	0.1790	0.2708	0.1599
RC2	RC2	Yes	0.9711	0.9590	0.9850	0.8199
RC2	RC1C2	No	0.7827	0.3215	0.4515	0.3380
RC1C2	R	No	0.9276	0.9189	0.9702	0.7211
RC1C2	C1	No	0.5662	0.6612	0.6012	0.2508
RC1C2	C2	No	0.9803	0.9777	0.9889	0.8503
RC1C2	RC1	No	0.6069	0.6833	0.6588	0.3208
RC1C2	RC2	No	0.9664	0.9506	0.9851	0.8122

Train	Test	In-dist	Precision	Recall	mAP50	mAP50-95
RC1C2	RC1C2	Yes	0.6807	0.7262	0.7489	0.4456

yolo26s

Train	Test	In-dist	Precision	Recall	mAP50	mAP50-95
R	R	Yes	0.9341	0.9097	0.9640	0.7074
R	C1	No	0.4140	0.0790	0.1334	0.0397
R	C2	No	0.9336	0.9158	0.9363	0.3638
R	RC1	No	0.5914	0.1738	0.2629	0.1531
R	RC2	No	0.9378	0.9099	0.9441	0.4795
R	RC1C2	No	0.7624	0.3034	0.4219	0.2027
RC1	R	No	0.9324	0.9392	0.9665	0.7000
RC1	C1	No	0.5759	0.6508	0.6264	0.2632
RC1	C2	No	0.9658	0.9158	0.9404	0.5529
RC1	RC1	Yes	0.6332	0.6636	0.6789	0.3305
RC1	RC2	No	0.9519	0.9237	0.9484	0.6127
RC1	RC1C2	No	0.7029	0.6860	0.7459	0.3858
RC2	R	No	0.9299	0.9291	0.9667	0.7175
RC2	C1	No	0.2241	0.1033	0.1508	0.0530
RC2	C2	No	0.9948	0.9824	0.9918	0.8669
RC2	RC1	No	0.6208	0.1660	0.2842	0.1707
RC2	RC2	Yes	0.9738	0.9613	0.9872	0.8247
RC2	RC1C2	No	0.8243	0.3032	0.4634	0.3499
RC1C2	R	No	0.9213	0.9054	0.9537	0.6948
RC1C2	C1	No	0.5897	0.6517	0.6291	0.2664
RC1C2	C2	No	0.9911	0.9811	0.9913	0.8599
RC1C2	RC1	No	0.6353	0.6692	0.6773	0.3314
RC1C2	RC2	No	0.9674	0.9556	0.9838	0.8129
RC1C2	RC1C2	Yes	0.7132	0.7068	0.7648	0.4584

yolo26m

Train	Test	In-dist	Precision	Recall	mAP50	mAP50-95
R	R	Yes	0.9415	0.9251	0.9620	0.6950
R	C1	No	0.0691	0.0452	0.0457	0.0107
R	C2	No	0.7409	0.7861	0.7993	0.3041
R	RC1	No	0.7230	0.1195	0.1955	0.1230
R	RC2	No	0.8159	0.8229	0.8712	0.4632
R	RC1C2	No	0.6781	0.2494	0.3170	0.1654
RC1	R	No	0.9048	0.9291	0.9579	0.6929
RC1	C1	No	0.5967	0.6708	0.6509	0.2702
RC1	C2	No	0.9308	0.9003	0.9450	0.4418
RC1	RC1	Yes	0.6418	0.6762	0.6952	0.3342
RC1	RC2	No	0.9225	0.9084	0.9411	0.5351
RC1	RC1C2	No	0.7020	0.6916	0.7552	0.3592
RC2	R	No	0.9101	0.9392	0.9582	0.7012
RC2	C1	No	0.1256	0.0928	0.1093	0.0362
RC2	C2	No	0.9913	0.9764	0.9929	0.8497
RC2	RC1	No	0.6265	0.1563	0.2388	0.1427
RC2	RC2	Yes	0.9710	0.9536	0.9850	0.8049
RC2	RC1C2	No	0.8219	0.2970	0.4237	0.3142
RC1C2	R	No	0.9181	0.9358	0.9590	0.6986
RC1C2	C1	No	0.6127	0.6246	0.6326	0.2654
RC1C2	C2	No	0.9844	0.9828	0.9921	0.8542
RC1C2	RC1	No	0.6615	0.6479	0.6823	0.3306
RC1C2	RC2	No	0.9603	0.9670	0.9854	0.8077
RC1C2	RC1C2	Yes	0.7381	0.6863	0.7671	0.4551